Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Value of x satisfying $\int_{\log 2}^{x} \frac{dx}{\sqrt{e^x-1}} = \frac{\pi}{6}$ is

(A) $\log 2$

(B) $3\log 2$

(C) $2\log 2$

(D) 2

Q.2

$\int_{-3\pi/2}^{-\pi/2} \left[(x+\pi)^3 + \cos^2(x+3\pi)\right] dx$ is equal to

(A) $(\pi/4) - 1$

(B) $\pi^4/32$

(C) $\left(\pi^4/32\right) + \pi/2$

(D) $\pi/2$

Q.3

$\int_0^1 \frac{dx}{(x^2+1)^{3/2}} =$

(A) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) 1

(D) $\sqrt{2}$

Q.4

If $\mathrm{I_a} = \int_0^{\pi/4} \tan^{\mathrm{n}}\theta \mathrm{d}\theta$, then $\mathrm{I_8} + \mathrm{I_6} =$

(A) $\frac{1}{4}$

(B) $\frac{1}{5}$

(C) $\frac{1}{6}$

(D) $\frac{1}{7}$

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Q.5

If $f(x)=\frac{e^x}{1+e^x}, I_1=\int_{f(-a)}^{f(a)} xg\{x(1-x)\}dx$ and $I_2=\int_{f(-a)}^{f(a)} g\{x(1-x)\}dx$, then the value of $\frac{I_2}{I_1}$ is

(A) 1

(B) -3

(C) -1

(D) 2

Q.6

$A_n=\int_0^{\pi/2}\frac{\sin(2n-1)x}{\sin x}dx; B_n=\int_0^{\pi/2}\left(\frac{\sin nx}{\sin x}\right)^2 dx$
For $\mathrm{n}\in\mathbf{N}$, then

(A) $A_{n+1}=A_n, B_{n+1}-B_n=A_{n+1}$

(B) $\mathrm{B_{n+1}=B_n}$

(C) $\mathrm{A_{n+1}-A_n=B_{n+1}}$

(D) None of these

Q.7

The value of $\int_0^1\frac{dx}{e^x+e}$ is equal to

(A) $\frac{1}{e}\log\left(\frac{1+e}{2}\right)$

(B) $\log\left(\frac{1+e}{2}\right)$

(C) $\frac{1}{e}\log(1+e)$

(D) $\log\left(\frac{2}{1+e}\right)$

Q.8

What is $\int_0^1\frac{\tan^{-1}}{1+\mathrm{x}^2}\mathrm{dx}$ equal to?

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{8}$

(C) $\frac{\pi^2}{8}$

(D) $\frac{\pi^2}{32}$

Q.9

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What is the value of $\int_0^1 xe^{x^2}dx$?

(A) $\frac{(e-1)}{2}$

(B) e^2-1

(C) $2(e-1)$

(D) $e-1$

Q.10

The value of $\int_0^{\pi}\ln(1+\cos x)dx$ is

(A) $\frac{\pi}{2}\log 2$

(B) $\pi\log 2$

(C) $-\pi(\log 2)$

(D) 0

Q.11

$\int_0^1 [f(x)g''(x)-f''(x)g(x)]\,dx$ is equal to :
[Given $f(0)=g(0)=0$]

(A) $f(1)g(1)-f(1)g'(1)$

(B) $f(1)g'(1)+f'(1)g(1)$

(C) $f(1)g'(1)-f'(1)g(1)$

(D) None of these

Q.12

What is the value of $\int_0^1(x-1)e^{-x}dx$?

(A) 0

(B) e

(C) $\frac{1}{e}$

(D) $\frac{-1}{e}$

Q.13

The value of
$\int_0^{\sin^2 x}\sin^{-1}\sqrt{t}dt+\int_0^{\cos^2 x}\cos^{-1}\sqrt{t}dt$ is

(A) π

(B) $\frac{\pi}{2}$

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(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{9}$

Q.14

If $\int_0^\infty \mathrm{e}^{-\mathrm{ax}}\mathrm{dx} = \frac{1}{\mathrm{a}}$, then $\int_0^\infty \mathrm{x}^{\mathrm{n}}\mathrm{e}^{-\mathrm{ax}}\mathrm{dx}$ is

(A) $\frac{(-1)^n n!}{a^{n+1}}$

(B) $\frac{(-1)^{\mathrm{n}}(\mathrm{n}-1)!}{\mathrm{a}^{\mathrm{n}}}$

(C) $\frac{n!}{a^{n+1}}$

(D) None of these

Q.15

What is the value of $\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$?

(A) $e\left(\frac{e}{2} - 1\right)$

(B) $\mathrm{e}(\mathrm{e} - 1)$

(C) $e - \frac{1}{e}$

(D) 0

Answers & Solutions

Q.1 Answer:

$2 \log 2$

Solution:

Substituting $e^x - 1 = t^2, e^x\, \mathrm{d}x = 2t$ **dt**

$\Rightarrow \quad \mathrm{d}x = \frac{2tdt}{t^2+1}, \ \mathrm{s}$

$$\frac{\pi}{6} = \int_{\log 2}^{x} \frac{\mathrm{d}x}{\sqrt{e^x - 1}} = \int_1^{\sqrt{e^x-1}} \frac{2tdt}{(t^2+1)\,t}$$

$$= 2\int_1^{\sqrt{e^x-1}} \frac{dt}{t^2+1} = 2\tan^{-1} t\Big|_1^{\sqrt{e^x-1}}$$

$$= 2\tan^{-1}\sqrt{e^x - 1} - \frac{\pi}{2}$$

$\Rightarrow \quad 2\tan^{-1}\sqrt{e^x - 1} = \frac{2\pi}{3}$

$\Rightarrow \quad \sqrt{e^x - 1} = \tan\frac{\pi}{3} = \sqrt{3}$

$\Rightarrow \quad e^x - 1 = 3 \quad$ **i.e.** $e^x = 4 \Rightarrow x = \log 4.$

$x = \log 2^2$

$= 2 \log 2$

Q.2 Answer:

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$\pi/2$

Solution:

$\int_{-3\pi/2}^{-\pi/2}\left[(x+\pi)^3+\cos^2(x+3\pi)\right]dx$

Let $x+\pi=t$

$\Rightarrow dx=dt$ **and limits are** $-\pi 2$ **and** $\pi 2$

$\Rightarrow \int_{-3\pi/2}^{-\pi/2}\left[t^3+\cos^2(2\pi+t]dx\right.$

$=2\int_0^{x/2}\cos^2 tdt=2\int_0^{x/2}\frac{1+\cos 2t}{2}dt=\frac{\pi}{2}$

Q.3 Answer:

$\frac{1}{\sqrt{2}}$

Solution:

Put $x=\tan\theta\Rightarrow \mathrm{dx}=\sec^2\theta \mathrm{d}\theta$

Also, when $x=0,\theta=0$ **and when** $x=1,\theta=\frac{\pi}{4}$

$\therefore I=\int_0^{\frac{\pi}{4}}\frac{1}{\sec^3\theta}\cdot\sec^2\theta d\theta$

$=\int_0^{\pi 4}\cos\theta d\theta=[\sin\theta]_0^{\pi/4}=\frac{1}{\sqrt{2}}$

Q.4 Answer:

$\frac{1}{7}$

Solution:

$\mathrm{I_g}+\mathrm{I_6}=\int_0^{\pi/4}\left(\tan^8\theta+\tan^6\theta\right)\mathrm{d}\theta$

$=\int_0^{\pi 4}\tan^6\theta\sec^2\theta d\theta=\left[\frac{\tan^7\theta}{7}\right]_0^{\pi/4}=\frac{1}{7}$

Q.5 Answer:

2

Solution:

$$f(x)=\frac{e^x}{1+e^x}\Rightarrow f(-x)=\frac{e^{-x}}{1+e^{-x}}$$
$$=\frac{1}{e^x+1}$$

$\therefore f(x)+f(-x)=1\forall x$ **Now**

$I_1=\int_{f(-a)}^{f}xg\{x(1-x)\}dx$

$=\int_{f(-a)}^{f}(1-x)g\{x(1-x)\}dx$

$=I_2-I_1\Rightarrow 2I_1=I_2$

Q.6 Answer:

$A_{n+1}=A_n, B_{n+1}-B_n=A_{n+1}$

Solution:

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We have

$A_{n+1} - A_n = \int_0^{\frac{\pi}{2}} \frac{\sin(2n+1)x - \sin(2n-1)x}{\sin x} dx$

$= \int_0^{\frac{\pi}{2}} \frac{2\cos 2nx \sin x}{\sin x} dx = 2\int_0^{\frac{\pi}{2}} \cos 2n\pi dx = 0$

Again $\mathrm{B_{n+1} - B_n} = \int_0^{\frac{\pi}{2}} \frac{\sin^2(\mathrm{n+1})\mathrm{x} - \sin^2 \mathrm{nx}}{\sin^2 \mathrm{x}} \mathrm{dx}$

$= \int_0^{\frac{\pi}{2}} \frac{\sin(2n+1)x \sin x}{\sin^2 x} dx = A_{n+1}$

Q.7 Answer:

$\frac{1}{e} \log\left(\frac{1+e}{2}\right)$

Solution:

$$\mathrm{Let I} = \int_0^1 \frac{dx}{e^x+e} = \int_0^1 \frac{e^x dx}{e^x(e^x+e)}$$
$$\text{put } e^x = t \Rightarrow e^x \mathrm{dx} = \mathrm{dt}$$
$$\mathrm{I} = \int_1^e \frac{dt}{t(t+e)} = \frac{1}{e}\int_1^e \left(\frac{1}{t} - \frac{1}{t+e}\right)$$
$$= \frac{1}{e}\int_1^e \frac{1}{t} dt - \frac{1}{e}\int_1^e \frac{1}{t+e} dt$$
$$= \frac{1}{e}[\log t]_1^e - \frac{1}{e}[\log(t+e)]_1^e$$
$$= \frac{1}{e}[\log t - \log(t+e)]_1^e$$
$$= \frac{1}{e}\left[\log\left(\frac{t}{t+e}\right)\right]_1^e = \frac{1}{e}\left[\log\left(\frac{e}{2e}\right) - \log\left(\frac{1}{1+e}\right)\right.$$
$$= \frac{1}{e}\log\left[\frac{1/2}{1/(1+e)}\right] = \frac{1}{e}\log\left(\frac{1+e}{2}\right)$$

Q.8 Answer:

$\frac{\pi^2}{32}$

Solution:

Let $I = \int_0^1 \frac{\tan^{-1}}{1+x^2} dx$

Put $\tan^{-1} x = t$

$\frac{1}{1+x^2} dx = dt$

$x = 0, \Rightarrow t = 0$

$x = 1, \Rightarrow t = \pi/4$

$\therefore \quad I = \int_0^{\pi/4} t dt = \left.\frac{t^2}{2}\right|_0^{\pi/4} = \frac{\pi^2}{32}$

Q.9 Answer:

$\frac{(\mathrm{e}-1)}{2}$

Solution:

Let $I = \int_0^1 x e^{x^2 dx}$

Let $x^2 = t$

$\Rightarrow 2\mathrm{xdx} = \mathrm{dt}$

$\Rightarrow \mathrm{xdx} = \frac{\mathrm{dt}}{2}$

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when $x=0, t=0$ then $x=1, t=1 \Rightarrow \mathrm{I}=\frac{1}{2}\int_0^1 \mathrm{e}^{\mathrm{t}}\mathrm{dt}=\frac{1}{2}\left[\mathrm{e}^{\mathrm{t}}\right]_0^1$
$=\frac{1}{2}\left[\mathrm{e}^{\mathrm{x}^2}\right]_0^1=\frac{1}{2}\left[\mathrm{e}-\mathrm{e}^0\right]=\frac{\mathrm{e}-1}{2}$

Q.10 Answer:

$-\pi(\log 2)$

Solution:

$I=\int_0^\pi \log(1+\cos x)dx=\int_0^\pi \log\left(2\cos^2\frac{x}{2}\right)dx$
$=\int_0^\pi\left(\log 2+2\log\cos\frac{x}{2}\right)dx$
$=\int_0^\pi \log 2\mathrm{dx}+2\int_0^\pi \log\cos\frac{x}{2}\mathrm{dx}$
$=\pi\log 2+2\int_0^{\pi/2}(2\log\cos t)\mathrm{dt}$ where $\frac{\mathrm{x}}{2}=\mathrm{t}$
$=\pi\log 2+4\left(-\frac{\pi}{2}\log 2\right)$
$=\pi\log 2-2\pi\log 2=-\pi\log 2$
$\left[\int_0^{\pi/2}\log\sin\theta\mathrm{d}\theta=\int_0^{\pi/2}\log\cos\theta\mathrm{d}\theta=-\frac{\pi}{2}\log 2\right.$

Q.11 Answer:

$\mathrm{f}(1)\mathrm{g}'(1)-f'(1)\mathrm{g}(1)$

Solution:

Integrating by parts. $\int \mathrm{f}(\mathrm{x})\mathrm{g}''(\mathrm{x})\mathrm{dx}-\int \mathrm{f}''(\mathrm{x})\mathrm{g}(\mathrm{x})\mathrm{dx}$
$=f(x)g'(x)-\int f'(x)g'(x)dx-f'(x)g(x)+\int f'(x)g'(x)dx$
$=f(x)g'(x)-f'(x)g(x)$
Hence, $\int_0^1 \mathrm{f}(\mathrm{x})\mathrm{g}''(\mathrm{x})\mathrm{dx}-\int_0^1 \mathrm{f}''(\mathrm{x})\mathrm{g}(\mathrm{x})\mathrm{dx}$
$=\mathrm{f}(1)\mathrm{g}'(1)-\mathrm{f}'(1)\mathrm{g}(1)-\mathrm{f}(0)\mathrm{g}'(0)+\mathrm{f}'(0)\mathrm{g}(0)$
$=\mathrm{f}(1)\mathrm{g}'(1)-\mathrm{f}'(1)\mathrm{g}(1)$

Q.12 Answer:

$\frac{-1}{\mathrm{e}}$

Solution:

Given integral is $I=\int_0^1(x-1)e^{-x}dx$ Integrating by parts taking $(\mathrm{x}-1)$ as first function We get, $I=\left[(x-1)\left\{-e^{-x}\right\}\right]_0^1-\int_0^1 1\cdot\left(-e^{-x}\right)dx$
$=-(1-1)\frac{1}{\mathrm{e}}+(-1)\mathrm{e}^0+\left[-\mathrm{e}^{-\mathrm{x}}\right]_0^1$
$=-1-\frac{1}{\mathrm{e}}+1=-\frac{1}{\mathrm{e}}$

Q.13 Answer:

$\frac{\pi}{4}$

Solution:

Let $I_1=\int_0^{\sin^2 x}\sin^{-1}\sqrt{t}dt$
Put $\mathrm{t}=\sin^2\mathrm{u}\Rightarrow \mathrm{dt}=2\sin\mathrm{u}\cos\mathrm{udu}$

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$\Rightarrow dt = \sin 2udu$

$\therefore I_1 = \int_0^x u \sin 2udu$

Let $I_2 = \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}dt$

Put $t = \cos^2 v \Rightarrow dt = -2\cos v \sin vdv$

$\Rightarrow dt = -\sin 2vdv$

$\therefore I_2 = \int_{\frac{\pi}{2}}^x v(-\sin 2v)dv = -\int_{\frac{\pi}{2}}^x v \sin 2vdv$

$= -\int_{\frac{\pi}{2}}^x u \sin 2udu$

[change of variable]

$\therefore I = I_1 + I_2 = \int_0^x u \sin 2udu - \int_{\frac{\pi}{2}}^x u \sin 2udu$

$= \int_0^{\frac{\pi}{2}} u \sin 2udu + \int_{\frac{\pi}{2}}^x u \sin 2udu - \int_{\frac{\pi}{2}}^x u \sin 2udu$

$= \int_0^{\frac{\pi}{2}} u \sin 2udu = \frac{\pi}{4}$ [**Integrate by parts**]

Q.14 Answer:

$\frac{n!}{a^{n+1}}$

Solution:

Let

$I_n = \int_0^\infty x^n e^{-ax} = \left[x^n \cdot \frac{e^{-ax}}{-a}\right]_0^\infty - \int_0^\infty nx^{n-1} \cdot \frac{e^{-ax}}{-a}dx$

$= -\frac{1}{a}\lim_{x\to\infty} \frac{x^n}{e^{ax}} + \frac{n}{a}I_{n-1} \quad \therefore I_n = \frac{n}{a}I_{n-1}$

$= \frac{n}{a} \cdot \frac{n-1}{a}I_{n-2} = \frac{n(n-1)(n-2)}{a^3}I_{n-3}$

-0

$= \frac{n!}{a^n}\int_0^\infty e^{-ax}dx = \frac{n!}{a^{n+1}}$

Q.15 Answer:

$e\left(\frac{e}{2} - 1\right)$

Solution:

Let $I = \int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$

$= \int_1^2 e^x \left(f(x) + f'(x)\right) dx$ **where** $f(x) = \frac{1}{x}$

$= e^x f(x)\Big|_1^2$

$\therefore \quad I = \frac{e^x}{x}\Big|_1^2 = \frac{e^2}{2} - e = e\left(\frac{e}{2} - 1\right)$

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